



SEGi
College

Subang Jaya

PROGRAMMING FUNDAMENTALS

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1.1 Chapter 1

1.1.1 /1.1.2 /1.1.3 The cout Object

Analysis

The first code is mainly to teach us how to output the code. Then, it does not require any input from the user because it is displaying a simple greeting message using the “cout” object. After we write out the code, it will display the message which is using “cout” function like “Hello, my name is Chew Jie Weei.” on the output screen.

Source Code

```
#include <iostream>

using namespace std;

int main()

{

    cout<< "Hello, my name is Chew Jie Weei.";

    cout<< "\n I am learning C++ using dev-C++ application";

    cout<< " Then C++ is one of these programming language\n other include python \n\n
    java and so on.";

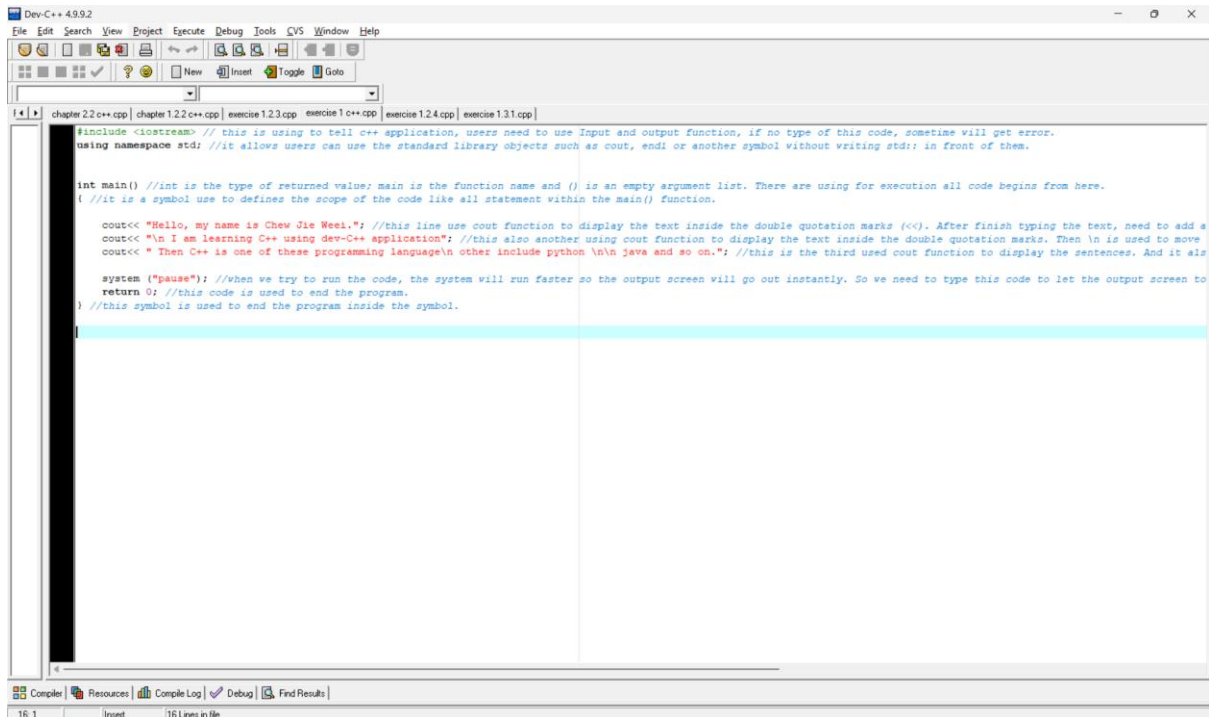
    system ("pause");

    return 0;

}
```

Screen shot

Source Code in DevC++



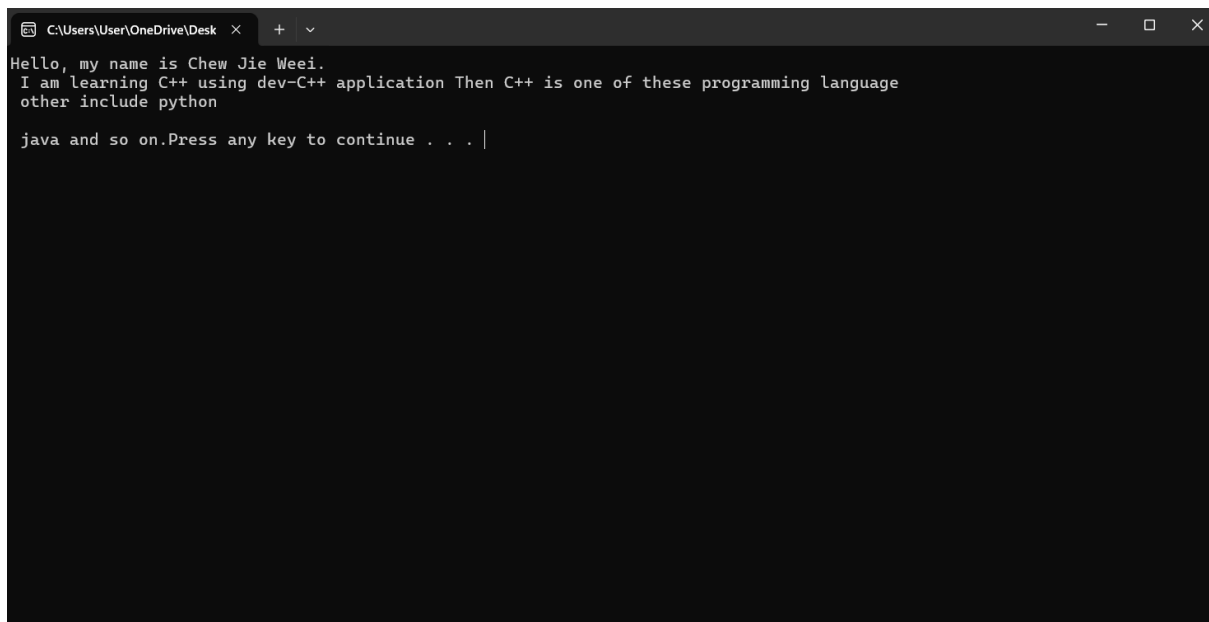
```
#include <iostream> // This is using to tell c++ application, users need to use Input and output function, if no type of this code, sometime will get error.
using namespace std; //it allows users can use the standard library objects such as cout, endl or another symbol without writing std:: in front of them.

int main() //int is the type of returned value; main is the function name and () is an empty argument list. There are using for execution all code begins from here.
{ //it is a symbol use to defines the scope of the code like all statement within the main() function.

    cout<< "Hello, my name is Chew Jie Weei."; //this line use cout function to display the text inside the double quotation marks (<<). After finish typing the text, need to add a
    cout<< "\n I am learning C++ using dev-C++ application"; //this also another using cout function to display the text inside the double quotation marks. Then \n is used to move
    cout<< " Then C++ is one of these programming language\n other include python \n\n java and so on."; //this is the third used cout function to display the sentences. And it als

    system("pause"); //when we try to run the code, the system will run faster so the output screen will go out instantly. So we need to type this code to let the output screen to
    return 0; //this code is used to end the program.
} //this symbol is used to end the program inside the symbol.
```

Output screen



```
C:\Users\User\OneDrive\Desk x + v
Hello, my name is Chew Jie Weei.
I am learning C++ using dev-C++ application Then C++ is one of these programming language
other include python
java and so on.Press any key to continue . . . |
```

1.2 Chapter 2

1.2.1 Data Type

Analysis

Data type is a set of values and operations that can be applied to these values. Common data types in C++ are int, float, double, char, string and bool.

Source Code

```
#include <iostream>

using namespace std;

int main ()
{
    //integer data types
    int totalAmount = 251;

    //character data types
    char section = 'A';

    //boolean data types
    bool isClassActive = true;


    //Floating-point data types
    float averageScore = 3.141;
    double Piperimeter = 3.14159265359;


    //Display all data types
    cout << "Total Amount:" << totalAmount << endl;
    cout << "Class Section:" << section << endl;
    cout << "Class Active:" << isClassActive << endl;
    cout << "Average Score:" << averageScore << endl;
    cout << "Pi Perimeter:" << Piperimeter << endl;
```

```

system("pause");

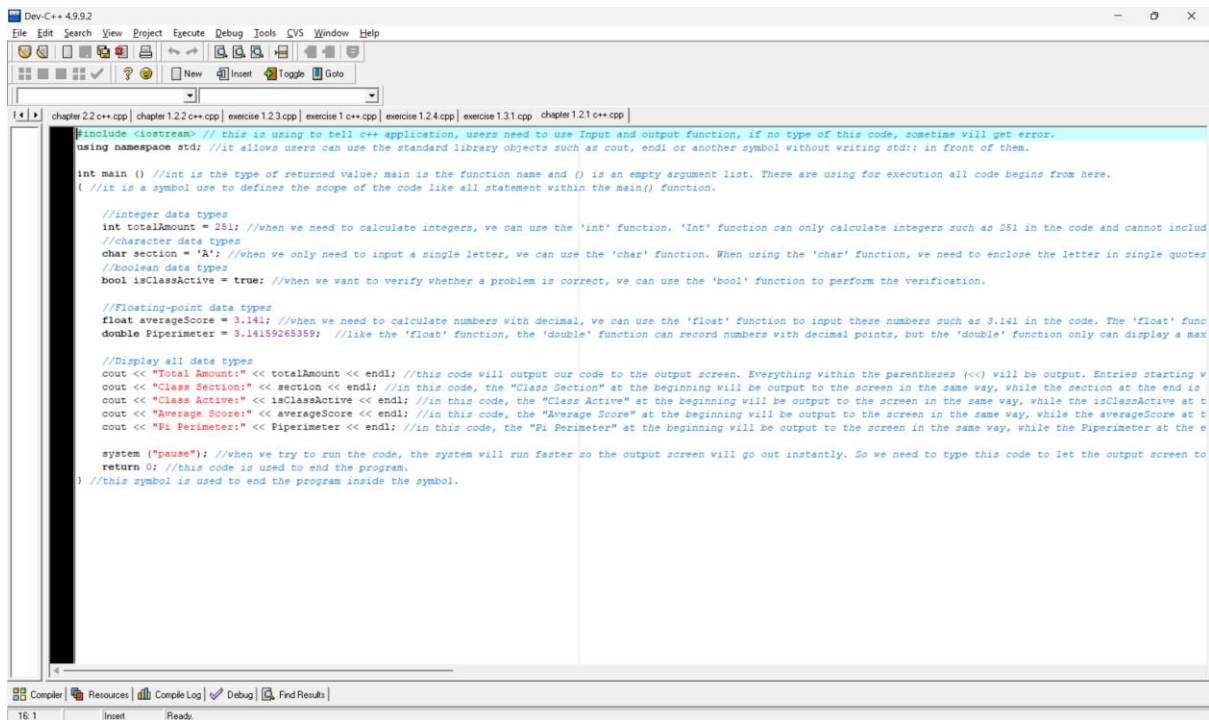
return 0;

}

```

Screen Shot

Source Code in DevC++



```

#include <iostream> // this is using to tell c++ application, users need to use input and output function, if no type of this code, sometime will get error.
using namespace std; //it allows users can use the standard library objects such as cout, endl or another symbol without writing std:: in front of them.

int main () //int is the type of returned value; main is the function name and () is an empty argument list. There are using for execution all code begins from here.
{ //it is a symbol use to defines the scope of the code like all statement within the main() function.

    //integer data types
    int totalAmount = 251; //when we need to calculate integers, we can use the 'int' function. 'Int' function can only calculate integers such as 251 in the code and cannot include

    //character data types
    char section = 'A'; //when we only need to input a single letter, we can use the 'char' function. When using the 'char' function, we need to enclose the letter in single quotes

    //boolean data types
    bool isClassActive = true; //when we want to verify whether a problem is correct, we can use the 'bool' function to perform the verification.

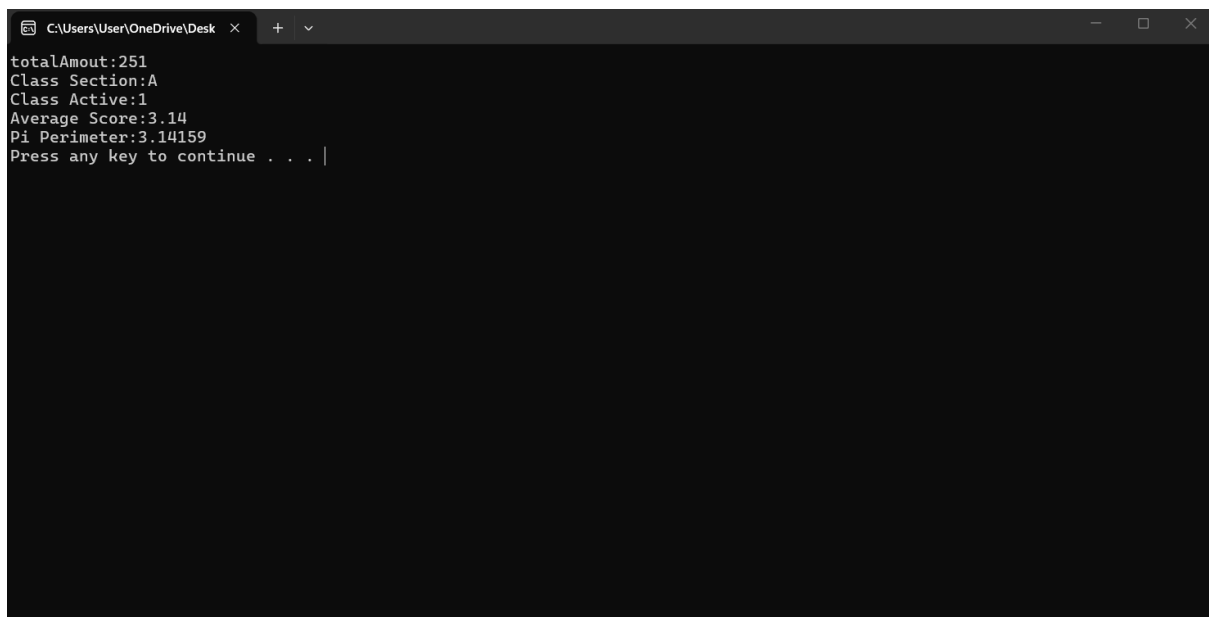
    //Floating-point data types
    float averageScore = 3.141; //when we need to calculate numbers with decimal, we can use the 'float' function to input these numbers such as 3.141 in the code. The 'float' function can record numbers with decimal points, but the 'double' function only can display a maximum of 15 decimal points.
    double PiPerimeter = 3.14159265359; //like the 'float' function, the 'double' function can record numbers with decimal points, but the 'double' function only can display a maximum of 15 decimal points.

    //Display all data types
    cout << "Total Amount:" << totalAmount << endl; //this code will output our code to the output screen. Everything within the parentheses (<<) will be output. Entries starting with cout << "Class Section:" << section << endl; //in this code, the "Class Section" at the beginning will be output to the screen in the same way, while the section at the end is cout << "Class Active:" << isClassActive << endl; //in this code, the "Class Active" at the beginning will be output to the screen in the same way, while the isClassActive at the end is cout << "Average Score:" << averageScore << endl; //in this code, the "Average Score" at the beginning will be output to the screen in the same way, while the averageScore at the end is cout << "Pi Perimeter:" << PiPerimeter << endl; //in this code, the "Pi Perimeter" at the beginning will be output to the screen in the same way, while the PiPerimeter at the end is

    system("pause"); //when we try to run the code, the system will run faster so the output screen will go out instantly. So we need to type this code to let the output screen to wait for the user to press a key.
    return 0; //this code is used to end the program.
} //this symbol is used to end the program inside the symbol.

```

Output Screen



```

totalAmount:251
Class Section:A
Class Active:1
Average Score:3.14
Pi Perimeter:3.14159
Press any key to continue . . . |

```

1.2.2 Declaration Statements

Analysis

A declaration statement is a statement used to tell the computer that I want to use a variable, what type the variable is and what its name is.

Source Code

```
#include <iostream>

using namespace std;

int main()
{
    //Declaration Statements

    int number;
    double price;
    char grade;
    bool passed;

    //Initialization (assigning values)

    number = 25;
    price = 125.35;
    grade = 'A';
    passed = true;

    //Display declared variables

    cout << "Number: " << number << endl;
    cout << "Price: RM " << price << endl;
    cout << "Grade: " << grade << endl;
    cout << "Passed: " << passed << endl;
```

```

system("pause");

return 0;

}

```

Screen Shot

Source Code in DevC++

```

#include <iostream> // this is using to tell c++ application, users need to use Input and output function, if no type of this code, sometime will get error.
using namespace std; //it allows users can use the standard library objects such as cout, endl or another symbol without writing std:: in front of them.

int main() //int is the type of returned value; main is the function name and () is an empty argument list. There are using for execution all code begins from here.
{ //it is a symbol use to defines the scope of the code like all statement within the main() function.

    //Declaration Statements
    int number; //used to store whole number
    double price; //like the 'float' function, the 'double' function can record numbers with decimal points, but the 'double' function only can display a maximum of 15 digits.
    char grade; //is used to store a single character such as a letter, pokemon or tiger word.
    bool passed; //is used to store 'True' or 'False' only.

    //Initialization (assigning values)
    number = 25; //is assigns 15 to the variable number.
    price = 125.35; //is assigns 125.35 to the variable price.
    grade = 'A'; // is assigns character 'A' for the grade.
    passed = true; //if the score is passed, it will display true and the output will change to 1.

    //Display declared variables
    cout << "Number: " << number << endl; //in this code, the "Number" at the beginning will be output to the screen in the same way, while the number at the end is the variable st
    cout << "Price: RM " << price << endl; //in this code, the "Price: RM" at the beginning will be output to the screen in the same way, while the price at the end is the variable
    cout << "Grade: " << grade << endl; //in this code, the "Grade" at the beginning will be output to the screen in the same way, while the grade at the end is the variable stored
    cout << "Passed: " << passed << endl; //in this code, the "Passed" at the beginning will be output to the screen in the same way, while the passed at the end is the variable st

    system("pause"); //when ve try to run the code, the system will run faster so the output screen will go out instantly. So we need to type this code to let the output screen to
    return 0; //this code is used to end the program.
} //this symbol is used to end the program inside the symbol.

```

Output screen

```

Number: 25
Price: RM 125.35
Grade: A
Passed: 1
Press any key to continue . . . |

```

1.2.3 Display

Analysis

‘Display’ is the system’s output of the code what the user input and ‘display’ can output variables of various types. Then the ‘Display’ output statement is preceded by the ‘cout’ function name.

Source Code

```
#include <iostream>

using namespace std;

int main()
{
    // Displaying text (string literals)
    cout << "Nice to meet you!" << endl;

    //Displaying numbers
    cout << "The minus of 15 and 7 is " << (17 - 5) << endl;

    //Displaying variables
    int num = 66;
    double time = 6.45;
    double price = 35;

    cout << "Number: " << num << endl;
    cout << "Time: " << time << endl;
    cout << "Price: " << price << endl;

    //using escape sequences
```

```
cout << "My father is a boy \n" ;
```

```
cout << "My mother is a girl \n" ;
```

```
//Multiple items in on cout statement
```

```
cout << "Total - " << num << ", Cost - RM " << price << ", So the time is " << time << endl;
```

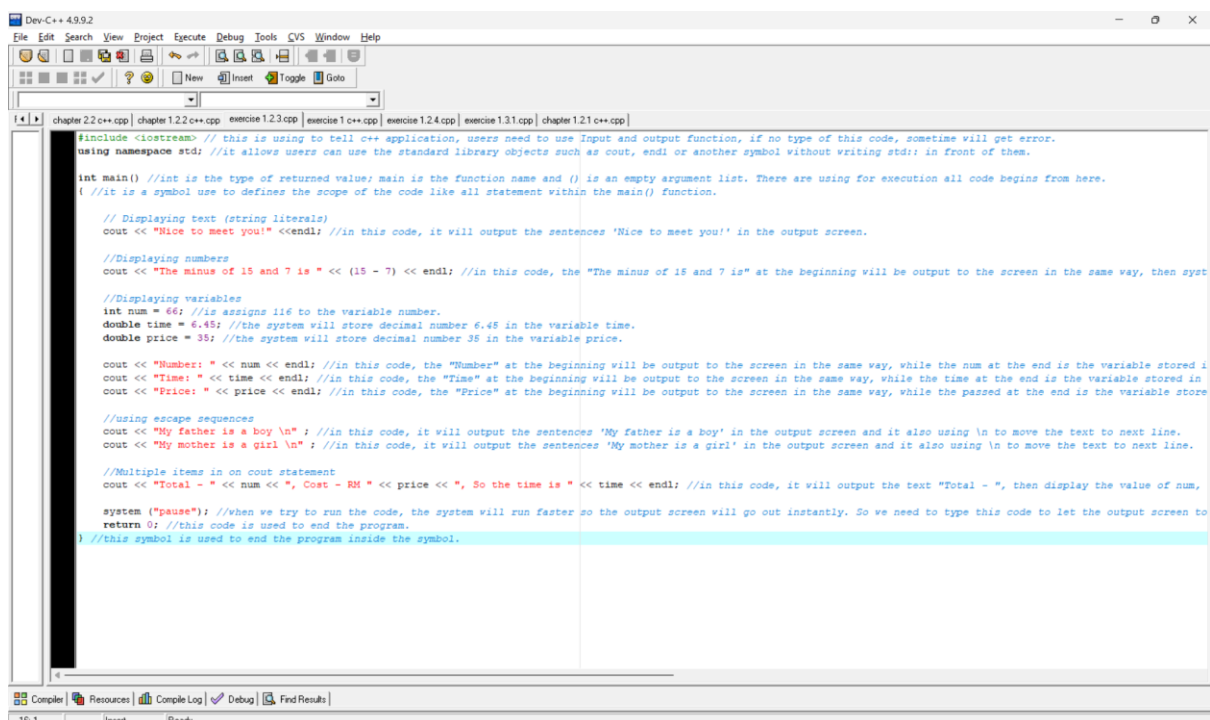
```
system ("pause");
```

```
return 0;
```

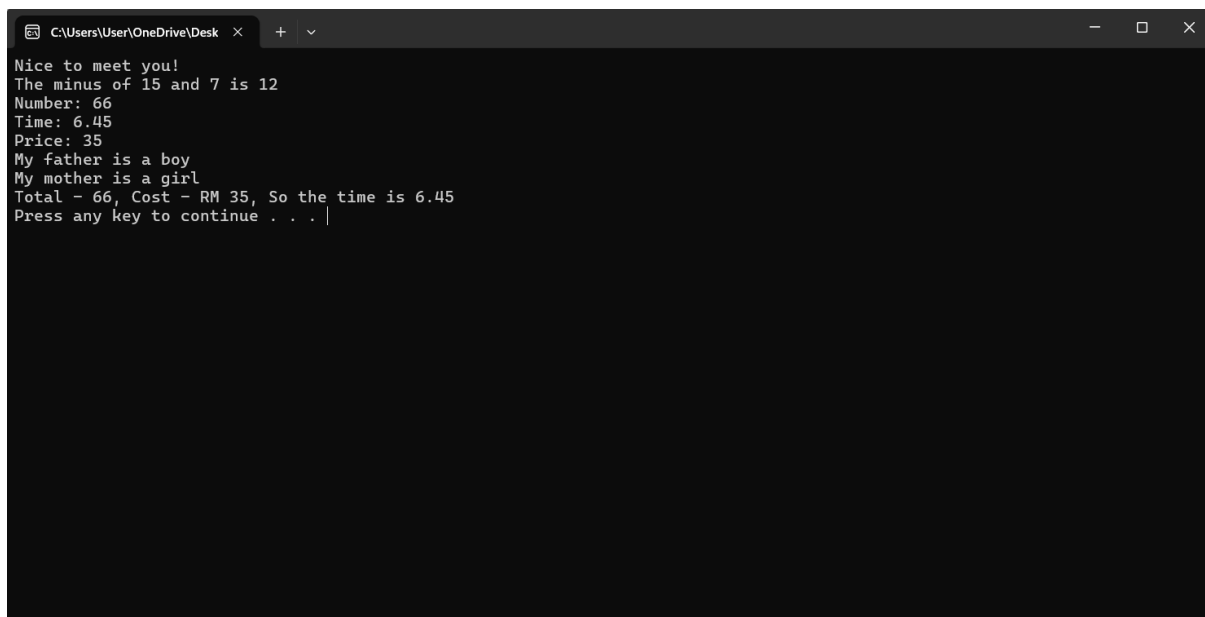
```
}
```

Screen Shot

Source Code in DevC++



Output Screen

A screenshot of a Windows command prompt window. The title bar shows the file path 'C:\Users\User\OneDrive\Desktop' and standard window controls. The command prompt displays the following text: 'Nice to meet you!', 'The minus of 15 and 7 is 12', 'Number: 66', 'Time: 6.45', 'Price: 35', 'My father is a boy', 'My mother is a girl', 'Total - 66, Cost - RM 35, So the time is 6.45', and 'Press any key to continue . . . |'.

```
C:\Users\User\OneDrive\Desktop >
Nice to meet you!
The minus of 15 and 7 is 12
Number: 66
Time: 6.45
Price: 35
My father is a boy
My mother is a girl
Total - 66, Cost - RM 35, So the time is 6.45
Press any key to continue . . . |
```

1.2.4 Declaration Statements

Analysis

A declaration statement is used to declare a variable in a program. It tells the system the data type of the variable and its name before it is used. Declaration allows the computer to reserve memory space for that variable. A declaration statement can declare variables of various data types such as integer, character, boolean and floating-point.

Source Code

```
#include <iostream>

using namespace std;

int main()
{
    char ch; //this declares a character variable

    ch = 'o' ; //store the word o in ch

    cout << "The character stored in ch is " << ch << endl;

    ch = 'f' ; //now stored the word f in ch
```

```

cout << "The character now stored in ch is" << ch << endl;

system ("pause");

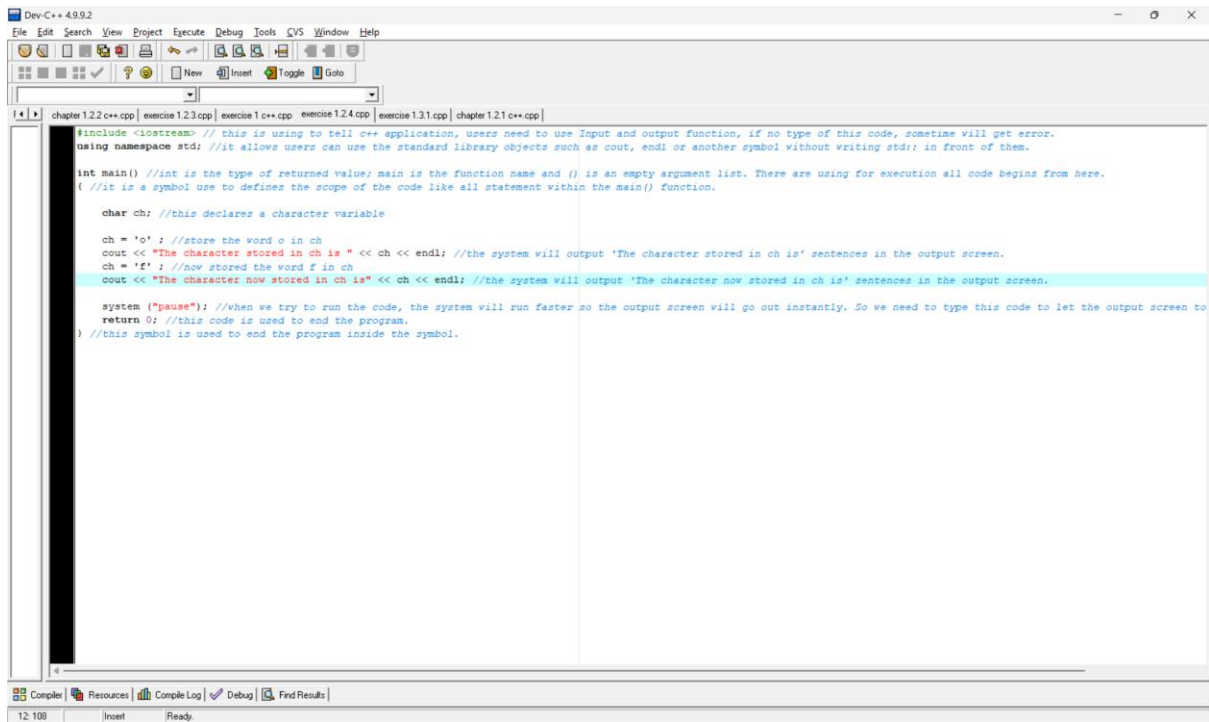
return 0;

}

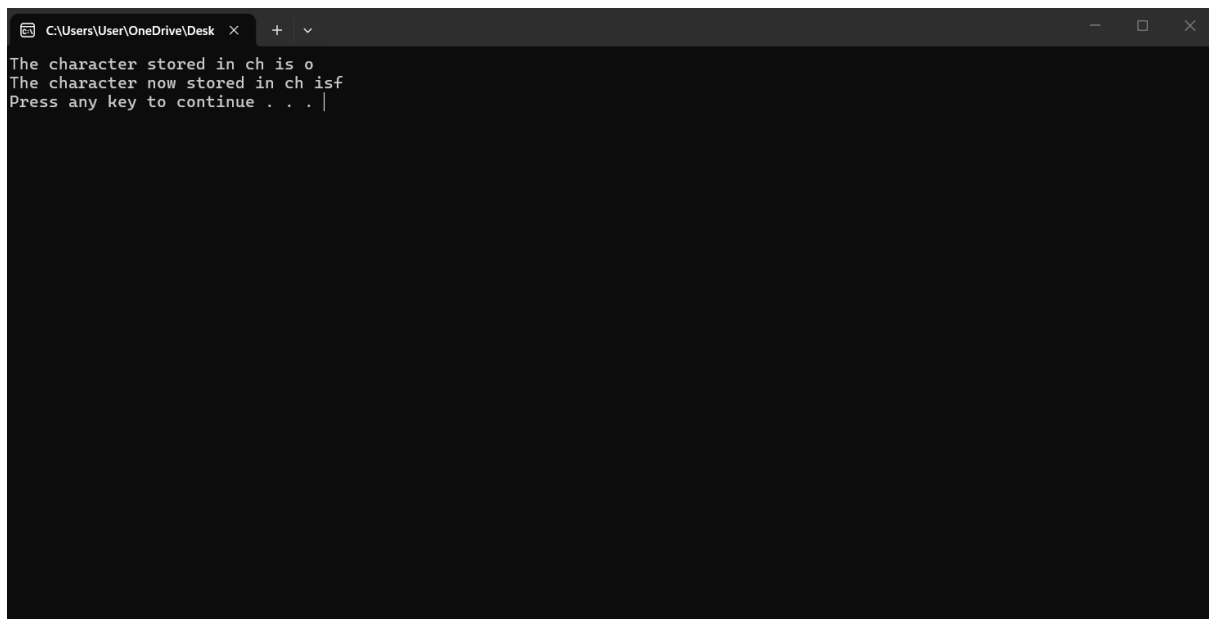
```

Screen Shot

Source Code in DevC++



Output Screen



1.3 Chapter 3

1.3.1 Assignment Operators

Analysis

‘Assignment Operators’ are used to assign values to variables in a program. The most common assignment operator is the equal sign (=), which stores the value on the right-hand side into the variable on the left-hand side.

Source Code

```
#include <iostream>

using namespace std;

int main()
{
    double length, width, area;

    length = 25;
    width = 15.25;
    area = length * width;

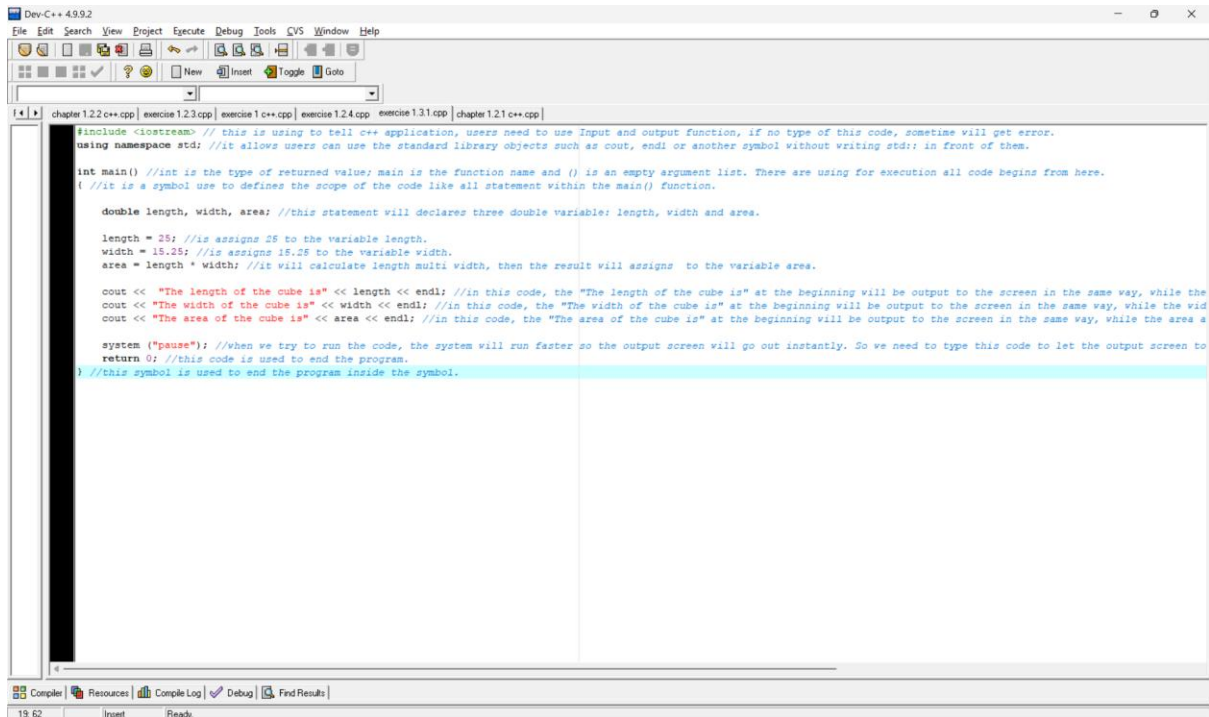
    cout << "The length of the cube is" << length << endl;
    cout << "The width of the cube is" << width << endl;
    cout << "The area of the cube is" << area << endl;

    system ("pause");

    return 0;
}
```

Screen Shot

Source Code in DevC++



```
#include <iostream> // this is using to tell c++ application, users need to use Input and output function, if no type of this code, sometime will get error.
using namespace std; //it allows users can use the standard library objects such as cout, endl or another symbol without writing std:: in front of them.

int main() //int is the type of returned value; main is the function name and () is an empty argument list. There are using for execution all code begins from here.
{ //it is a symbol use to defines the scope of the code like all statement within the main() function.

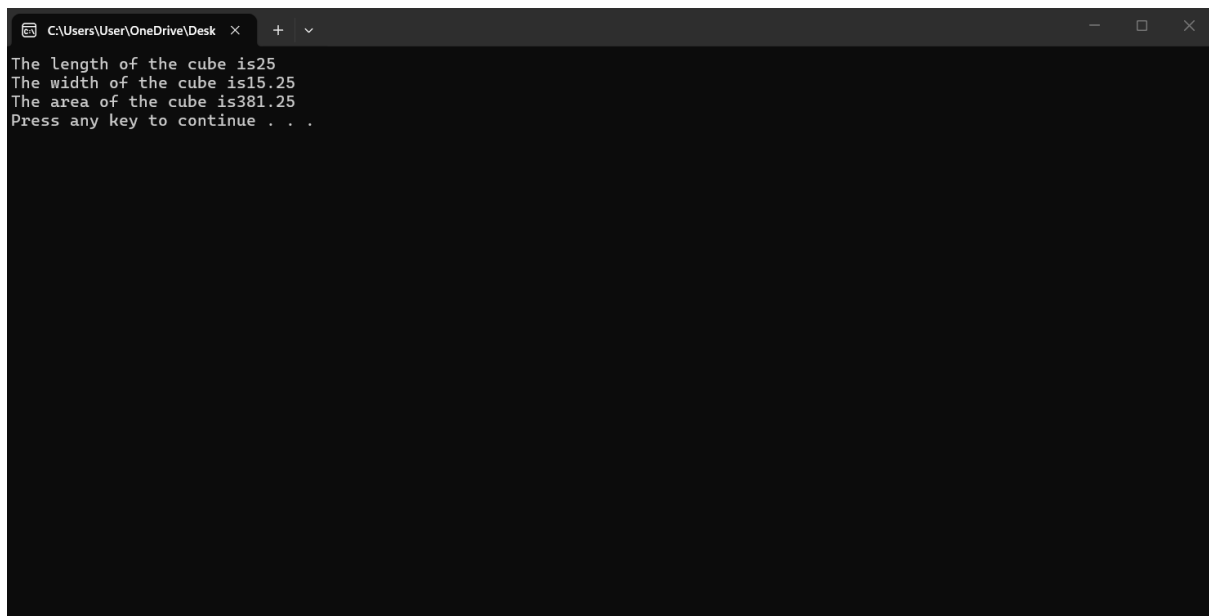
    double length, width, area; //this statement will declares three double variable: length, width and area.

    length = 25; //is assigns 25 to the variable length.
    width = 15.25; //is assigns 15.25 to the variable width.
    area = length * width; //it will calculate length multi width, then the result will assigns to the variable area.

    cout << "The length of the cube is" << length << endl; //in this code, the "The length of the cube is" at the beginning will be output to the screen in the same way, while the
    cout << "The width of the cube is" << width << endl; //in this code, the "The width of the cube is" at the beginning will be output to the screen in the same way, while the vid
    cout << "The area of the cube is" << area << endl; //in this code, the "The area of the cube is" at the beginning will be output to the screen in the same way, while the area a

    system("pause"); //when we try to run the code, the system will run faster so the output screen will go out instantly. So we need to type this code to let the output screen to
    return 0; //this code is used to end the program.
} //this symbol is used to end the program inside the symbol.
```

Output Screen



```
C:\Users\User\OneDrive\Desktop >
The length of the cube is25
The width of the cube is15.25
The area of the cube is381.25
Press any key to continue . . .
```

1.3.2 Assignment Variations

Analysis

Assignment Variations is another way to assign values to a variable and it can be written by using the following shortcut operators like +=, -=, *=, /= and %=.

Source Code

```
#include <iostream>

using namespace std;

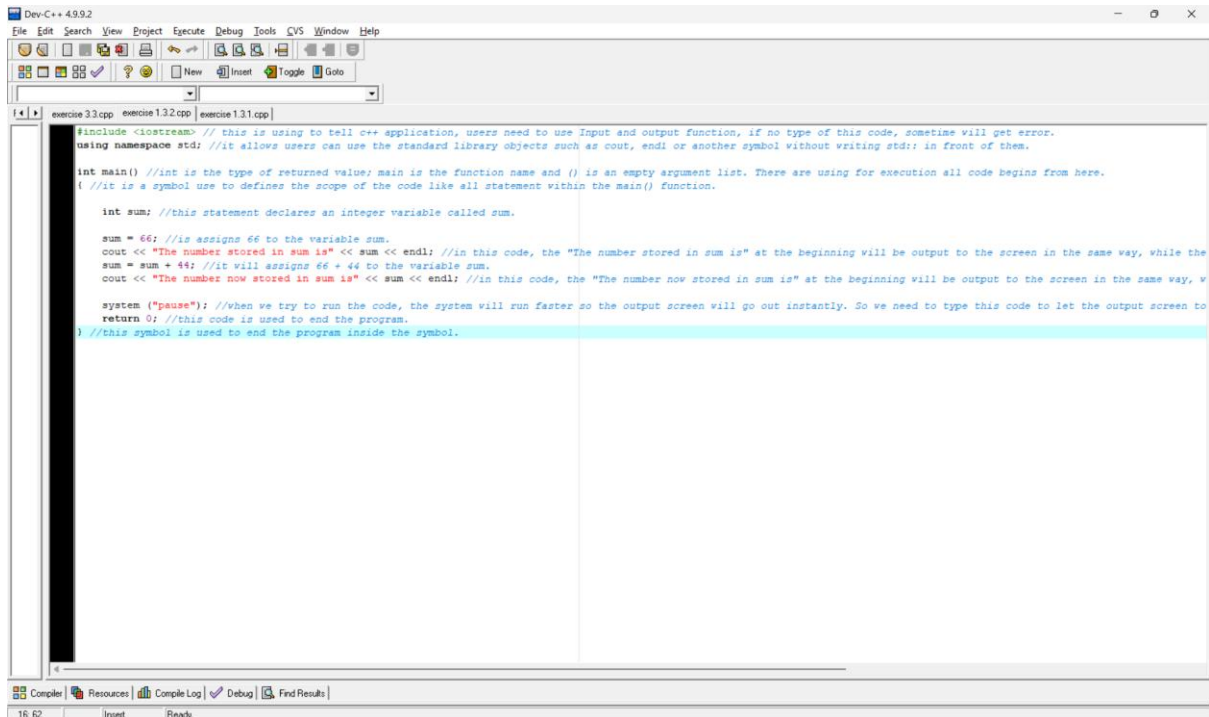
int main()
{
    int sum;

    sum = 66;
    cout << "The number stored in sum is" << sum << endl;
    sum = sum + 44;
    cout << "The number now stored in sum is" << sum << endl;

    system ("pause");
    return 0;
}
```

Screen Shot

Source Code in DevC++



```
#include <iostream> // this is using to tell c++ application, users need to use Input and output function, if no type of this code, sometime will get error.
using namespace std; //it allows users can use the standard library objects such as cout, endl or another symbol without writing std:: in front of them.

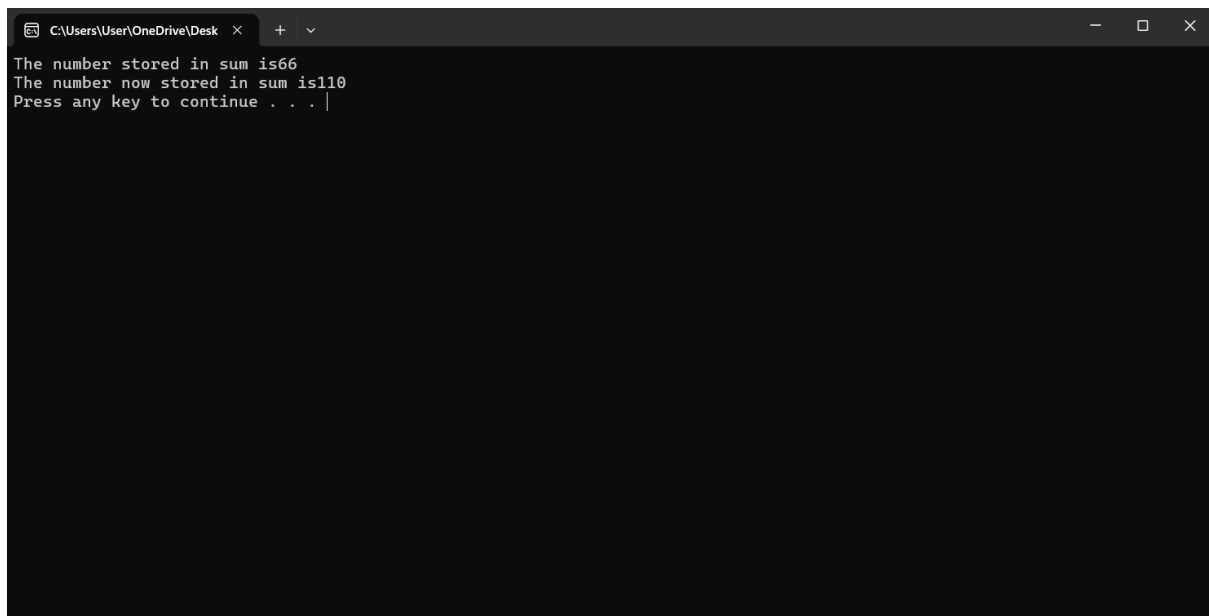
int main() //int is the type of returned value; main is the function name and () is an empty argument list. There are using for execution all code begins from here.
{ //it is a symbol use to defines the scope of the code like all statement within the main() function.

    int sum; //this statement declares an integer variable called sum.

    sum = 66; //is assigns 66 to the variable sum.
    cout << "The number stored in sum is" << sum << endl; //in this code, the "The number stored in sum is" at the beginning will be output to the screen in the same way, while the
    sum = sum + 44; //it will assigns 66 + 44 to the variable sum.
    cout << "The number now stored in sum is" << sum << endl; //in this code, the "The number now stored in sum is" at the beginning will be output to the screen in the same way, v

    system("pause"); //when we try to run the code, the system will run faster so the output screen will go out instantly. So we need to type this code to let the output screen to
    return 0; //this code is used to end the program.
} //this symbol is used to end the program inside the symbol.
```

Output Screen



```
C:\Users\User\OneDrive\Desk x + v
The number stored in sum is66
The number now stored in sum is110
Press any key to continue . . . |
```

1.3.3 Accumulating

Analysis

Accumulating is the process of repeatedly adding values to a variable and storing the updated result back into the same variable.

Source Code

```
#include <iostream>

using namespace std;

int main()
{
    int sum;

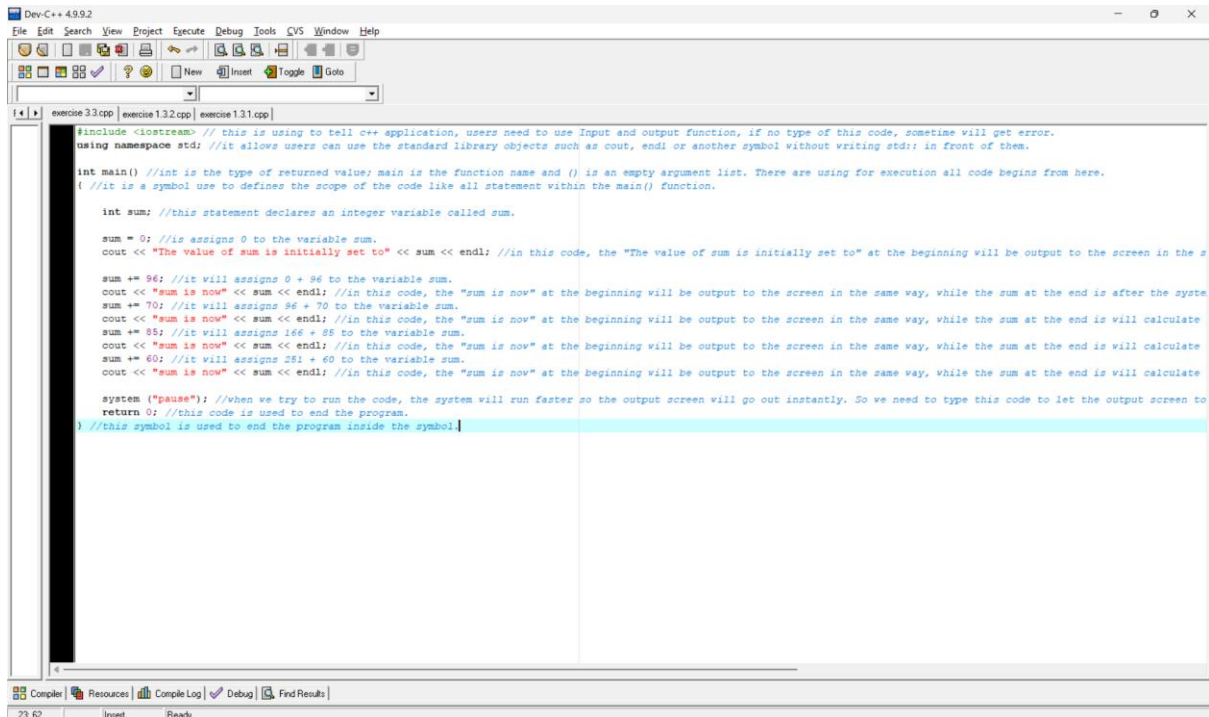
    sum = 0;
    cout << "The value of sum is initially set to" << sum << endl;

    sum += 96;
    cout << "sum is now" << sum << endl;
    sum += 70;
    cout << "sum is now" << sum << endl;
    sum += 85;
    cout << "sum is now" << sum << endl;
    sum += 60;
    cout << "sum is now" << sum << endl;

    system("pause");*
    return 0;
}
```

Screen Shot

Source Code in DevC++



```
#include <iostream> // this is using to tell c++ application, users need to use Input and output function, if no type of this code, sometime will get error.
using namespace std; //it allows users can use the standard library objects such as cout, endl or another symbol without writing std:: in front of them.

int main() //int is the type of returned value; main is the function name and () is an empty argument list. There are using for execution all code begins from here.
{ //it is a symbol use to defines the scope of the code like all statement within the main() function.

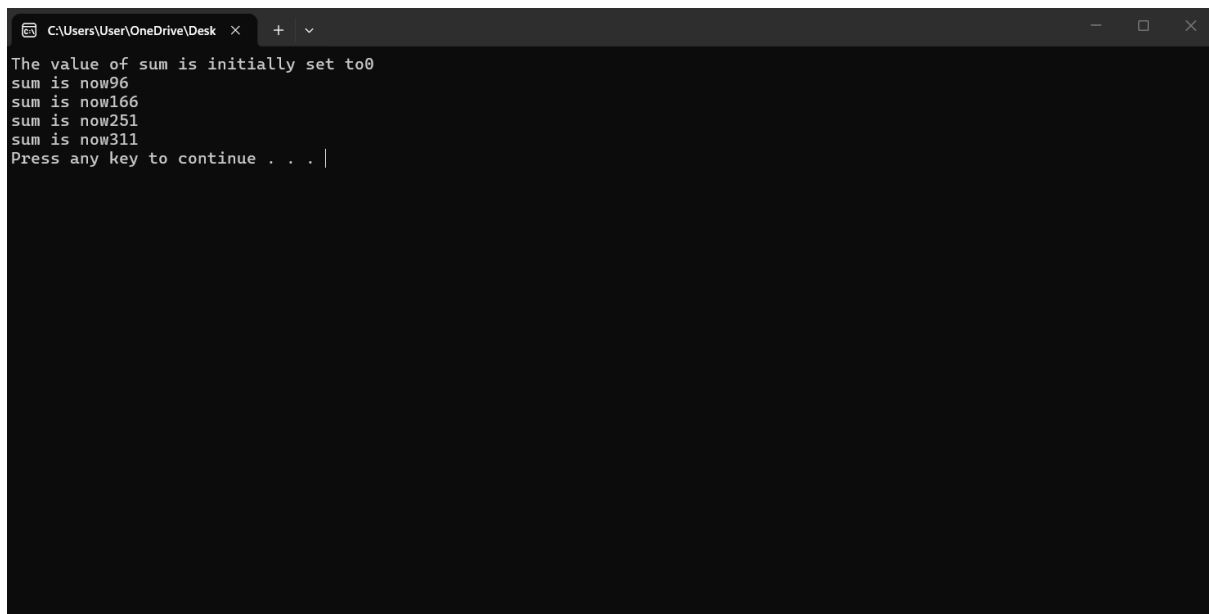
    int sum; //this statement declares an integer variable called sum.

    sum = 0; //is assigns 0 to the variable sum.
    cout << "The value of sum is initially set to" << sum << endl; //in this code, the "The value of sum is initially set to" at the beginning will be output to the screen in the s

    sum += 96; //it will assigns 0 + 96 to the variable sum.
    cout << "sum is now" << sum << endl; //in this code, the "sum is now" at the beginning will be output to the screen in the same way, while the sum at the end is after the syste
    sum += 70; //it will assigns 96 + 70 to the variable sum.
    cout << "sum is now" << sum << endl; //in this code, the "sum is now" at the beginning will be output to the screen in the same way, while the sum at the end is will calculate
    sum += 85; //it will assigns 166 + 85 to the variable sum.
    cout << "sum is now" << sum << endl; //in this code, the "sum is now" at the beginning will be output to the screen in the same way, while the sum at the end is will calculate
    sum += 60; //it will assigns 251 + 60 to the variable sum.
    cout << "sum is now" << sum << endl; //in this code, the "sum is now" at the beginning will be output to the screen in the same way, while the sum at the end is will calculate

    system("pause"); //when we try to run the code, the system will run faster so the output screen will go out instantly. So we need to type this code to let the output screen to
    return 0; //this code is used to end the program.
} //this symbol is used to end the program inside the symbol. }
```

Output Screen



```
C:\Users\User\OneDrive\Desktop x + v
The value of sum is initially set to0
sum is now96
sum is now166
sum is now251
sum is now311
Press any key to continue . . . |
```

1.3.4 Counting

Analysis

Counting is the process of increasing a variable by 1 each time an event occurs and it also have another function called decrement counting and it is decreasing a counter variable by 1 each time an event occurs.

Source Code

```
#include <iostream>

using namespace std;

int main()
{
    int count;

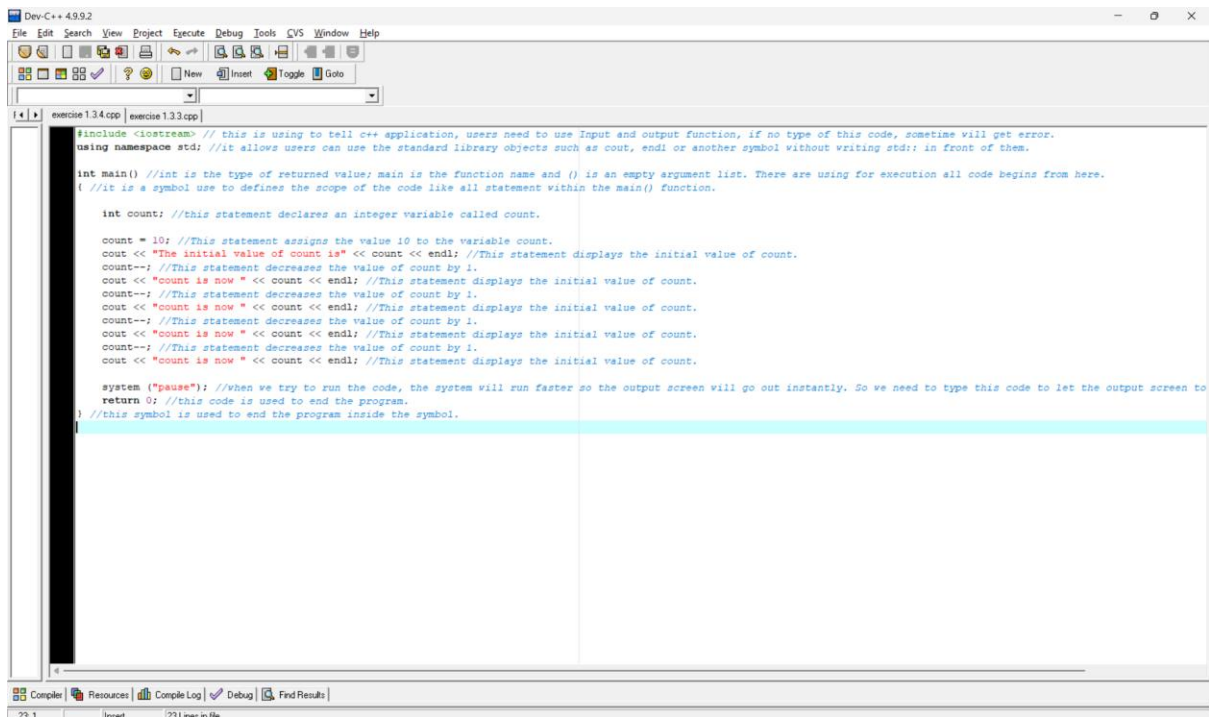
    count = 10;
    cout << "The initial value of count is" << count << endl;
    count--;
    cout << "count is now " << count << endl;
    count--;
    cout << "count is now " << count << endl;
    count--;
    cout << "count is now " << count << endl;
    count--;
    cout << "count is now " << count << endl;

    system ("pause");

    return 0;
}
```

Screen Shot

Source Code in DevC++



```
#include <iostream> // this is using to tell c++ application, users need to use Input and output function, if no type of this code, sometime will get error.
using namespace std; //it allows users can use the standard library objects such as cout, endl or another symbol without writing std:: in front of them.

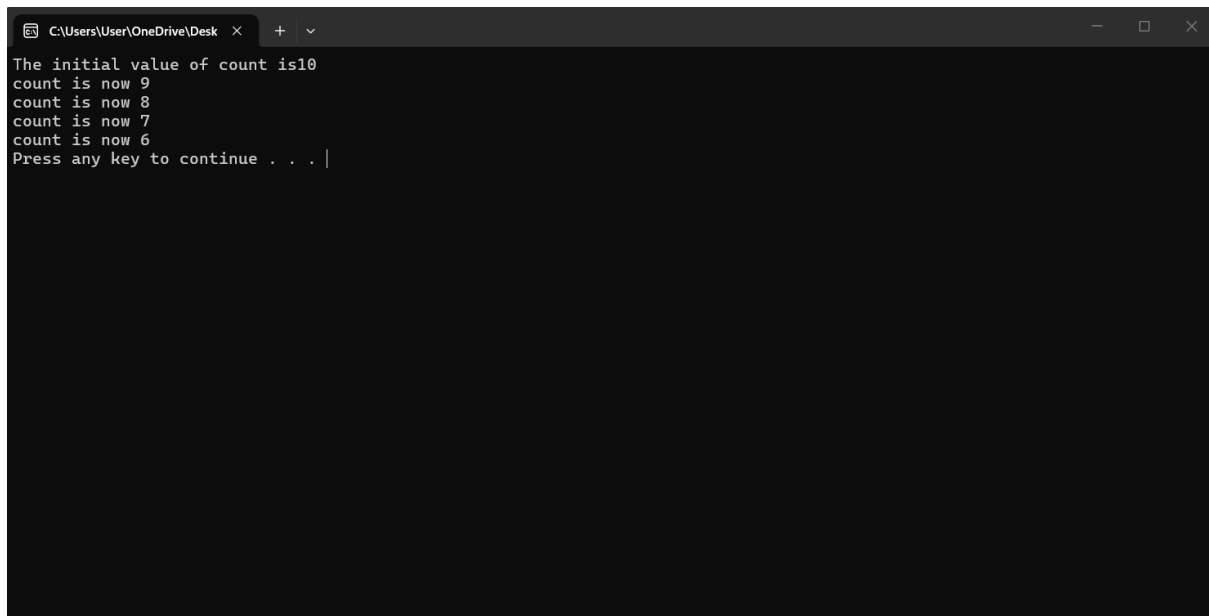
int main() //int is the type of returned value; main is the function name and () is an empty argument list. There are using for execution all code begins from here.
{ //it is a symbol use to defines the scope of the code like all statement within the main() function.

    int count; //this statement declares an integer variable called count.

    count = 10; //This statement assigns the value 10 to the variable count.
    cout << "The initial value of count is" << count << endl; //This statement displays the initial value of count.
    count--; //This statement decreases the value of count by 1.
    cout << "count is now " << count << endl; //This statement displays the initial value of count.
    count--; //This statement decreases the value of count by 1.
    cout << "count is now " << count << endl; //This statement displays the initial value of count.
    count--; //This statement decreases the value of count by 1.
    cout << "count is now " << count << endl; //This statement displays the initial value of count.
    count--; //This statement decreases the value of count by 1.
    cout << "count is now " << count << endl; //This statement displays the initial value of count.

    system("pause"); //when we try to run the code, the system will run faster so the output screen will go out instantly. So we need to type this code to let the output screen to
    return 0; //this code is used to end the program.
} //this symbol is used to end the program inside the symbol.
```

Output Screen



```
C:\Users\User\OneDrive\Desktop >
The initial value of count is10
count is now 9
count is now 8
count is now 7
count is now 6
Press any key to continue . . . |
```

1.3.5 Formatted Output

Analysis

Formatted Output is a program used to present results attractively and it use field width manipulators function to control the format of number displayed by cout. Then the manipulators are included in the output stream.

Source Code

```
#include <iostream>

using namespace std;

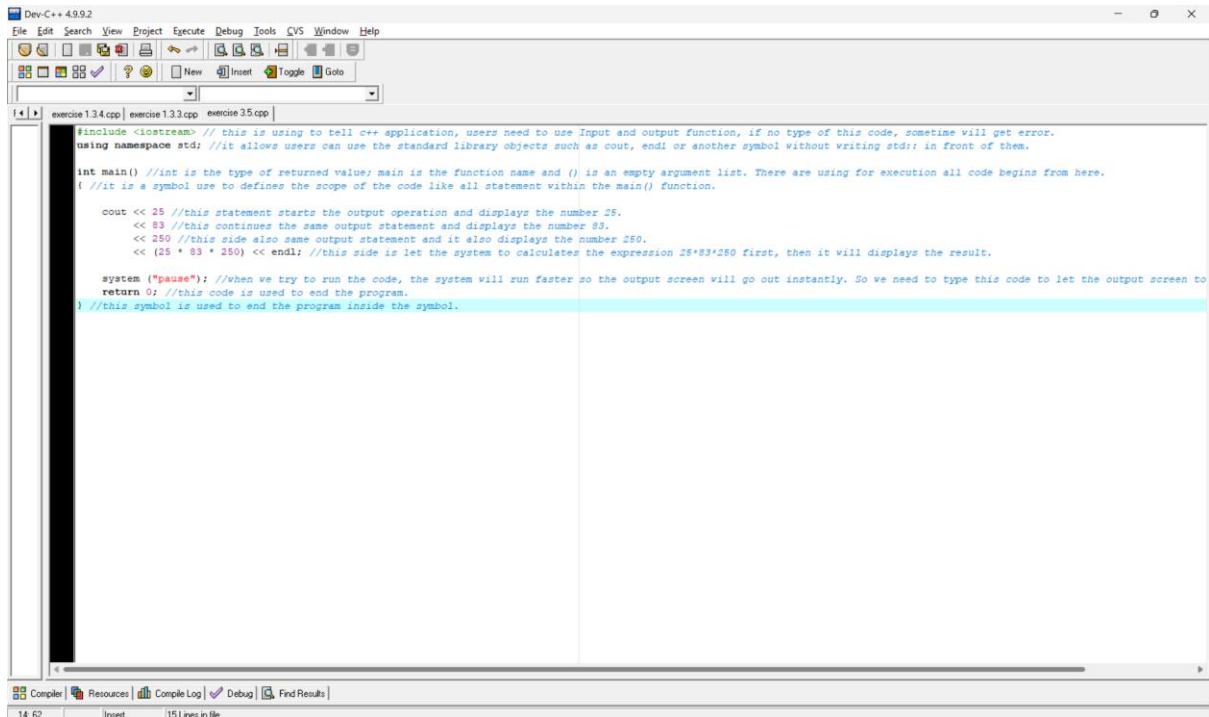
int main()
{

    cout << 25
        << 83
        << 250
        << (25 * 83 * 250) << endl;

    system ("pause");
    return 0;
}
```

Screen Shot

Source Code in DevC++



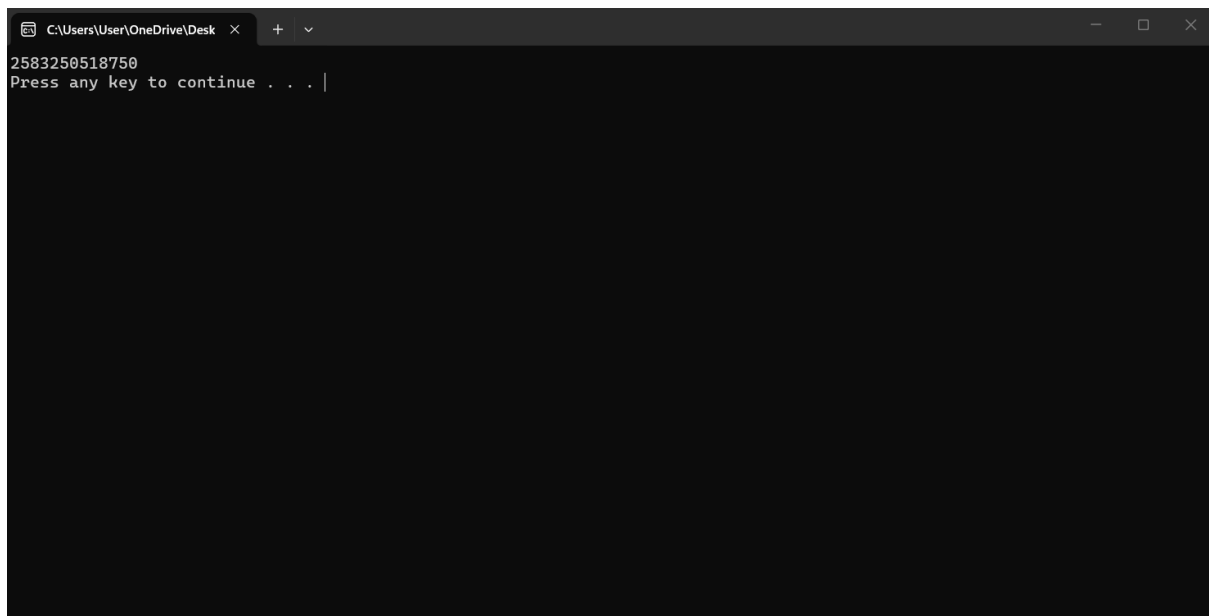
```
#include <iostream> // this is using to tell c++ application, users need to use Input and output function, if no type of this code, sometime will get error.
using namespace std; //it allows users can use the standard library objects such as cout, endl or another symbol without writing std:: in front of them.

int main() //int is the type of returned value; main is the function name and () is an empty argument list. There are using for execution all code begins from here.
{ //it is a symbol use to defines the scope of the code like all statement within the main() function.

    cout << 25 //this statement starts the output operation and displays the number 25.
    << 83 //this continues the same output statement and displays the number 83.
    << 250 //this side also same output statement and it also displays the number 250.
    << (25 * 83 * 250) << endl; //this side is let the system to calculates the expression 25*83*250 first, then it will displays the result.

    system("pause"); //when we try to run the code, the system will run faster so the output screen will go out instantly. So we need to type this code to let the output screen to
    return 0; //this code is used to end the program.
} //this symbol is used to end the program inside the symbol.
```

Output Screen



```
C:\Users\User\OneDrive\Desktop x + v
2583250518750
Press any key to continue . . . |
```

1.3.6 Hexadecimal and Octal I/O

Analysis

Hexadecimal and Octal manipulators are used for conversions to octal and hexadecimal. It will display of integer values in one of three possible numbering systems (decimal, octal and hexadecimal) doesn't affect how the number is stored in a computer and all the numbers are stored by using the computer's internal codes.

Source Code

```
#include <iostream>

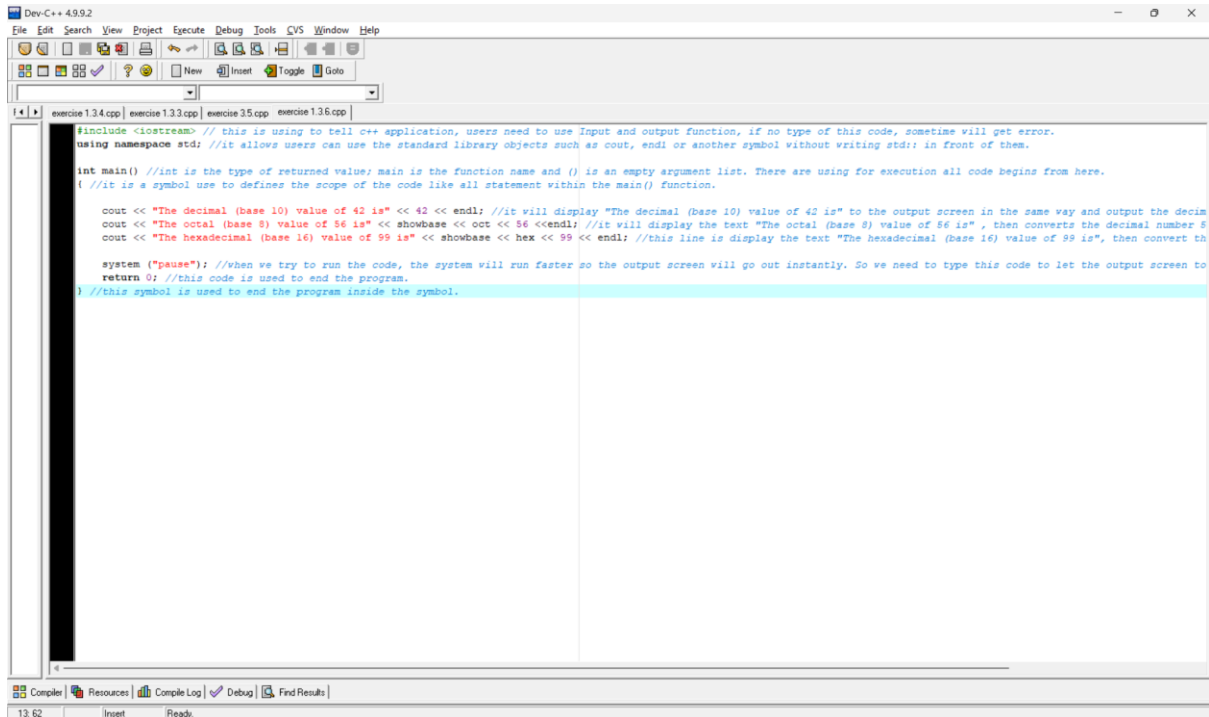
using namespace std;

int main()
{
    cout << "The decimal (base 10) value of 42 is" << 42 << endl;
    cout << "The octal (base 8) value of 56 is" << showbase << oct << 56 << endl;
    cout << "The hexadecimal (base 16) value of 99 is" << showbase << hex << 99 << endl;

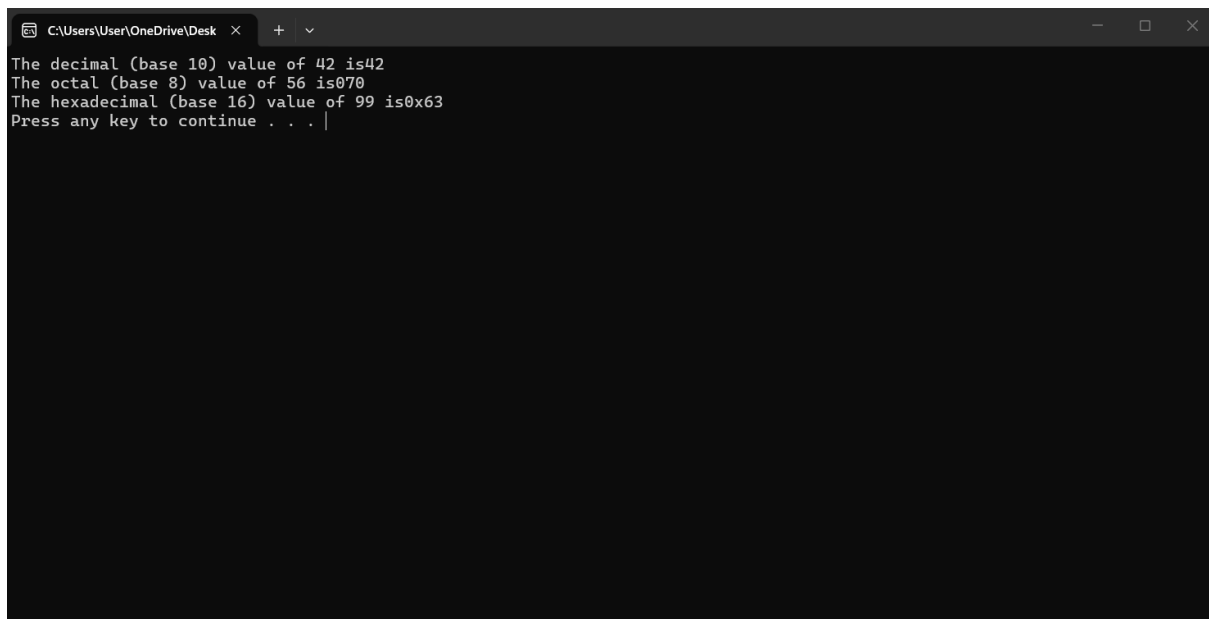
    system ("pause");
    return 0;
}
```

Screen Shot

Source Code in DevC++



Output Screen



1.3.7 Mathematical Library Functions

Analysis

Mathematical Library Functions is a standard preprogrammed functions that can be included in a program such as sqrt(x), sin(x), cos(x) and tan(x). If need to use these functions in a program, the header file <cmath> must be used.

Source Code

```
#include <iostream>

#include <cmath>

using namespace std;

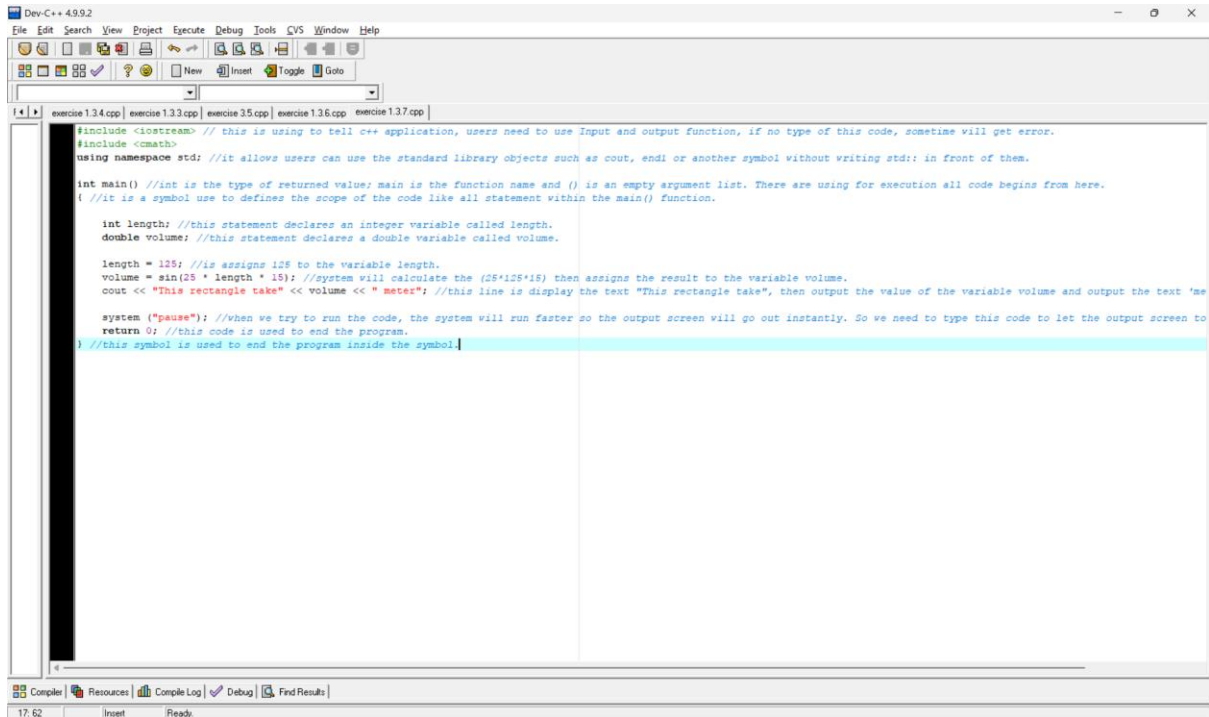
int main()
{
    int length;
    double volume;

    length = 125;
    volume = sin(25 * length * 15);
    cout << "This rectangle take" << volume << " meter";

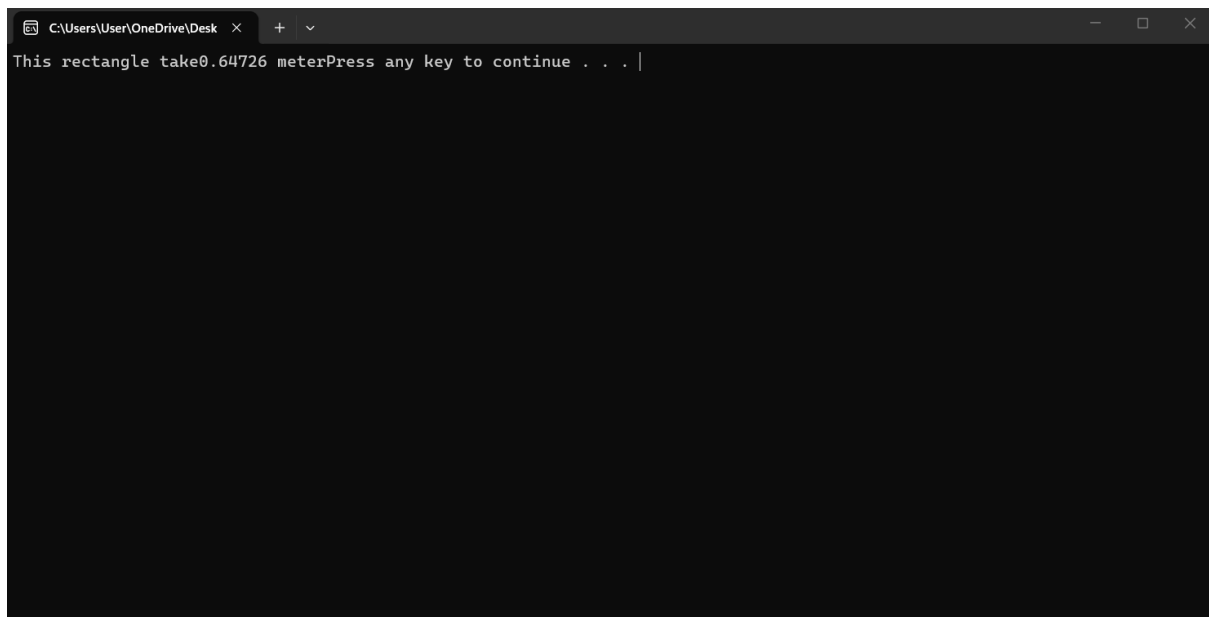
    system ("pause");
    return 0;
}
```

Screen Shot

Source Code in DevC++



Output Screen



1.3.8 Interactive Keyboard Input

Analysis

Interactive keyboard input is used to let the user to input data from the keyboard during runtime and then reading and processing that data. If a program only executes once, data can be included in the program. If using this function, need to add 'cin' object to let the user enter the data while the program is executing.

Source Code

```
#include <iostream>

using namespace std;

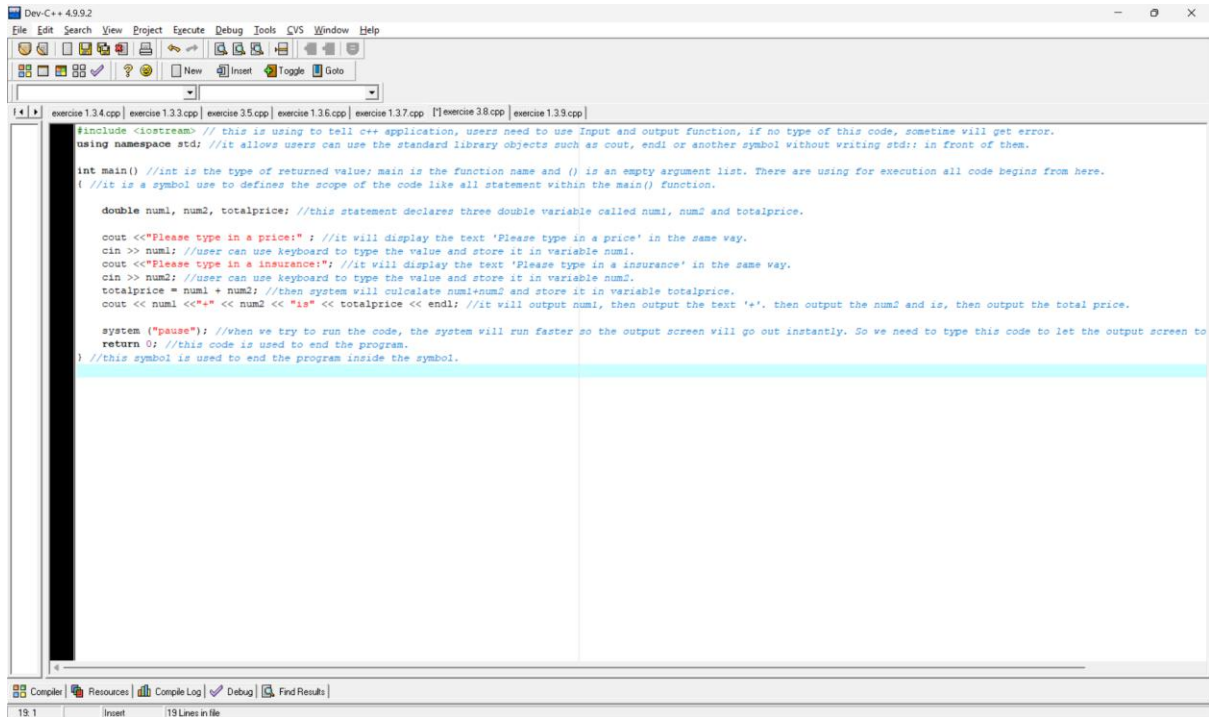
int main()
{
    double num1, num2, totalprice;

    cout << "Please type in a price:" ;
    cin >> num1;
    cout << "Please type in a insurance:";
    cin >> num2;
    totalprice = num1 + num2;
    cout << num1 << "price" << num2 << "is" << totalprice << endl;

    system ("pause");
    return 0;
}
```

Screen Shot

Source Code in DevC++



```
Dev-C++ 4.9.9.2
File Edit Search View Project Execute Debug Tools CVS Window Help

#include <iostream> // this is using to tell c++ application, users need to use Input and output function, if no type of this code, sometime will get error.
using namespace std; //it allows users can use the standard library objects such as cout, endl or another symbol without writing std:: in front of them.

int main() //int is the type of returned value; main is the function name and () is an empty argument list. There are using for execution all code begins from here.
{ //it is a symbol use to defines the scope of the code like all statement within the main() function.

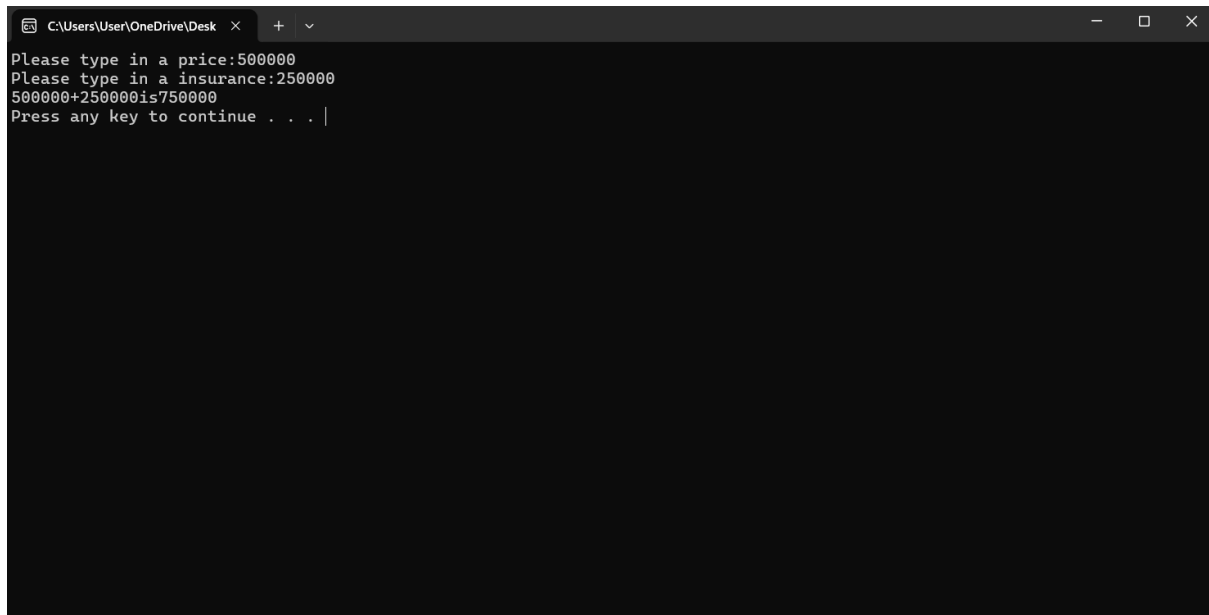
    double num1, num2, totalPrice; //this statement declares three double variable called num1, num2 and totalPrice.

    cout << "Please type in a price:" ; //it will display the text 'Please type in a price' in the same way.
    cin >> num1; //user can use keyboard to type the value and store it in variable num1.
    cout << "Please type in a insurance:"; //it will display the text 'Please type in a insurance' in the same way.
    cin >> num2; //user can use keyboard to type the value and store it in variable num2.
    totalPrice = num1 + num2; //then system will calculate num1+num2 and store it in variable totalPrice.
    cout << num1 << "+" << num2 << "is" << totalPrice << endl; //it will output the num1, then output the text '+', then output the num2 and is, then output the total price.

    system("pause"); //when we try to run the code, the system will run faster so the output screen will go out instantly. So we need to type this code to let the output screen to
    return 0; //this code is used to end the program.
} //this symbol is used to end the program inside the symbol.

Compiler | Resources | Compile Log | Debug | Find Results |
19:1 | Insert | 19 Lines in file
```

Output Screen



```
C:\Users\User\OneDrive\Desktop >
Please type in a price:500000
Please type in a insurance:250000
500000+250000is750000
Press any key to continue . . . |
```

1.3.9 Placement of Statements

Analysis

Statement placement is a function that refer to the location or order of statements in a program. The order of statements affects how the program executes because C++ executes statements sequentially from top to bottom. Improper statement placement can lead to errors or incorrect results. To ensure correct program execution, variables must be declared before use, input must be obtained before computation and results must be output after computation.

Source Code

```
#include <iostream>

#include <iomanip>

using namespace std;

int main()
{

    const int NUM_SUBJECTS = 3;
    double score1, score2, score3, average;

    cout << "Enter the score for subject 1: ";
    cin >> score1;
    cout << "Enter the score for subject 2: ";
    cin >> score2;
    cout << "Enter the score for subject 3: ";
    cin >> score3;

    average = (score1 + score2 + score3) / NUM_SUBJECTS;
```

```

cout << setiosflags (ios::fixed)

    << setiosflags (ios::showpoint)

    << setprecision(2);

cout << "Score 1: " << setw(5) << score1 << endl;

cout << "Score 2: " << setw(5) << score2 << endl;

cout << "Score 3: " << setw(5) << score3 << endl;

cout << "Average : " << setw(5) << average << endl;

system ("pause");

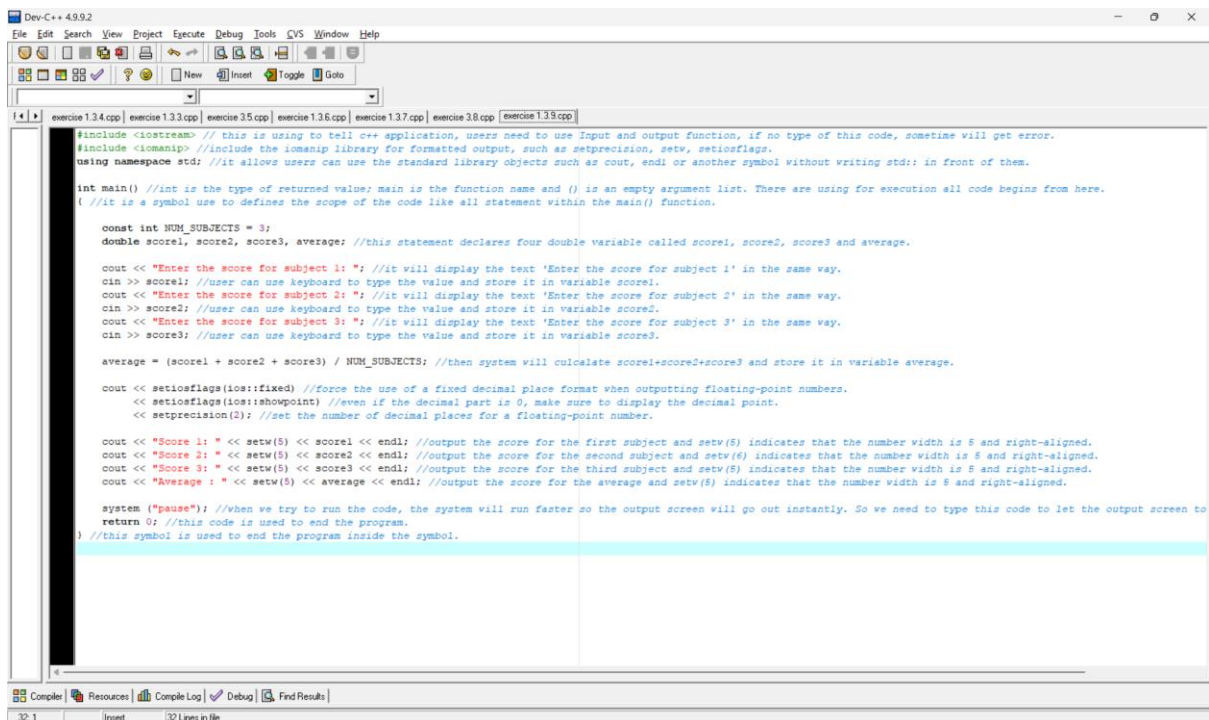
return 0;

}

```

Screen Shot

Source Code in DevC++



Output Screen

```
C:\Users\User\OneDrive\Desk x + v
Enter the score for subject 1: 60
Enter the score for subject 2: 70
Enter the score for subject 3: 80
Score 1: 60.00
Score 2: 70.00
Score 3: 80.00
Average : 70.00
Press any key to continue . . . |
```